on two of the most important clusters in the history of this topic: Governance and Blockchain. As shown in Fig. 1(c) and in Fig. S7 in the SM, these two clusters are among the biggest across time in terms of number of papers. The Governance cluster has always been first or second with respect to the other clusters at the 3rd level, while Blockchain was barely present before 2008, the year of the bitcoin white paper [4]. After that, Blockchain gradually increased in size and had an exponential explosion after 2015, coincidentally with the increasing hype around the technology and its applications, in particular bitcoin and ethereum [30, 31, 32]. Finally, it has become the most productive cluster since 2019, surpassing governance.

To better understand their role in the evolution of the literature on *(de)centralization*, we consider the average knowledge flows between clusters for each year, that is looking at $K_{a\to\bullet}(Y)$, $K_{\bullet\to a}(Y)$, and $K_{\bullet\to\bullet}(Y)$, defined in Eq. (5). Therefore, in Fig. 3 we rank clusters year by year using $K_{a\to\bullet}(Y)$ in (a) and $K_{\bullet\to a}(Y)$ in (b), i.e., looking at how much the papers of a cluster a in a year Y have influenced, on average, the future of all other clusters (a), or, vice versa, how much all clusters have influenced the future of a (b). From these plots we can see how, on the one hand, Governance has been in the top ranks until the late 1980s, both as a source and target of knowledge flows. However, in the early 1990s it started to decrease in importance, reaching the bottom ranks in the 2000s, despite being the first cluster in terms of number of papers each of these years. On the other hand, in Fig. 3(a) we notice that the rise of Blockchain started only in 2010, being almost always outside of the *(de)centralization* literature discussion until this point. Then, very sharply, Blockchain becomes the first cluster in terms of influence towards other clusters in 2013, maintaining its position in the following years. Hence, the literature on Blockchain has been key in the development of the *(de)centralization* discussion in the most recent years. Moreover, looking at Blockchain in Fig. 3(b), papers of other clusters before early 2000s have had almost no impact on the scientific future of Blockchain. Interestingly, it has received a lot of influence from publications between 2006 and 2012, that is about when the blockchain and bitcoin originated [4], as well as after 2017, mostly due to the increasing amount of applications using blockchain in the most diverse contexts in recent years. Finally, notice the loss of influence on Blockchain from papers between 2013 and 2016.

These results are corroborated by the time evolution of the average knowledge flow compared to the overall average $K_{\bullet\to\bullet}(Y)$. Indeed, in Fig. S8 in the SM we show how Governance has been increasingly important in influencing other clusters until the 1980s, while since the 1990s it has had a lower average knowledge flow than the average among all clusters. Similarly to what shown by the ranks, after 2013 Blockchain starts to have a much higher influence towards the other clusters compared to the average. Moreover, in Fig S9 in the SM we compare the average knowledge flow within the same cluster and towards other clusters, isolating the clusters Governance and Blockchain from the rest of the clusters. We find that on the one hand Governance has maintained a very high knowledge flow to future papers in the same cluster throughout the years. On the other hand, starting from 1990s, the exchange in knowledge flow towards and from other clusters has decreased. Blockchain, instead, has received a higher than average knowledge flow from other clusters in the 2000s, starting to provide maximum knowledge flow to itself and more than average to other clusters in the next and most recent decade.

So far we have seen how Governance has been influential in the early literature about *(de)centralization*, and how Blockchain has risen in recent years as the most important influential cluster, contributing in terms of knowledge flow towards other branches of literature. It is therefore a natural next step to investigate with higher granularity which clusters in particular have influenced or have been influenced

by Governance first and Blockchain then, and see how these interactions have changed over time. We start this analysis from the more recent case of Blockchain. This cluster started to appear only around 2008 with the bitcoin white paper [4]. Moreover, we notice a decrease in the influence on this cluster in mid 2010s. We therefore divide the 2008–2020 time span in three parts, following the blockchain history: 2008–2014, representing the origin of blockchain applications before the advent of ethereum; 2015–2018, when the field got more recognition thanks to ethereum and bitcoin; and the final 2019–2020 period, in which we have seen the explosion of academic literature production and the widespread success of multiple applications such as DeFi, NFTs and the metaverse.

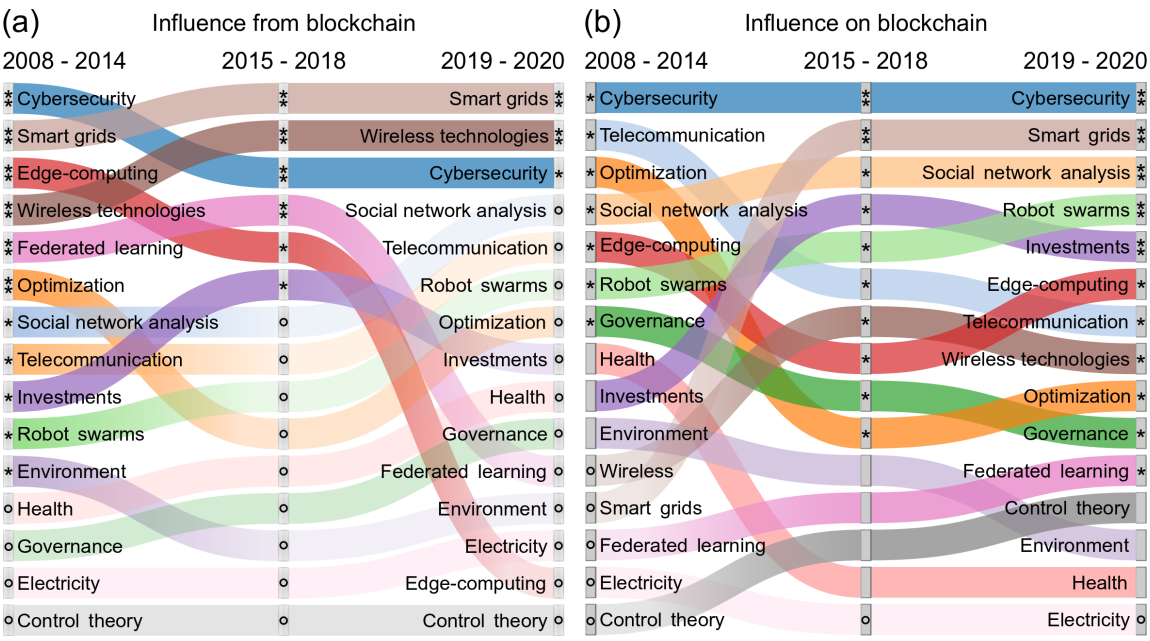

Figure 4: **Influences between Blockchain and the other clusters on *(de)centralization*. (a)** Change in the ranking of the most influenced clusters by Blockchain between its early period (2008-2014), its middle period (2015-2018), and its late period (2019-2020), calculated using the average knowledge flows $K_{a\to b}(T)$, where T is the selected period, and a is fixed to be Blockchain. **(b)** Change in the ranking of the clusters having most influenced the Blockchain literature (same periods as in the previous panel). In both cases, if $K_{a\to b}(T) = 0$, we print a circle in the corresponding gray node and use a lighter color in the respective link. Moreover, we print a star when $0.01 < K_{a\to b}(T) \leq 0.1$, and two stars when $K_{a\to b}(T) > 0.1$.

In Fig. 4 we plot, in a decreasing order, which clusters have been most influenced by (a) and have most influenced (b) Blockchain during the three periods. To this end, we use a Sankey diagram, showing how the overall picture has changed in the three different phases. The plot is done using the average knowledge flows $K_{a\to b}(T)$, where T is the selected period, while a and b are fixed to Blockchain in Fig. 4(a) and Fig. 4(b) respectively.

Firstly, in Fig. 4(a) we analyze the influence of Blockchain on the these other clusters. The early literature of Blockchain has had a strong impact on most of the clusters. As a matter of fact, there are only a few cases where the average knowledge flow from Blockchain to another cluster is zero, shown by a circle in the respective node and a lighter color in the corresponding link. We also notice that Cybersecurity, Smart grids, Edge-computing, Wireless technologies, and Federated learning have a very significant average knowledge flow from Blockchain, i.e., $K_{a\to b}(T) > 0.1$, shown by the double stars, while other clusters with $0.01 < K_{a\to b}(T) \leq 0.1$ are represented with only one stars. Notice how Blockchain

has continued to have a big impact on these mentioned clusters, in particular Smart grids and Wireless technologies, as well as Cybersecurity to a lesser extent. Looking altogether at the three periods, notice how Cybersecurity and Edge-computing have lost influence from Blockchain over time. Moreover, in the last period there is no significant knowledge flow to all other clusters, which is peculiar if we consider that, for example, Federated learning and Edge-computing received a very significant knowledge flow in the previous years. We argue that the recent decrease in knowledge flow is mostly due to the time needed for a paper to attract citations, especially outside its own cluster. Additionally, we find that some clusters, such as Health, Electricity, Control theory and Governance, have received no significant influence from Blockchain in all these years, even if, Governance, for instance, has been second only to Blockchain in terms of number of papers.

Secondly, in Fig. 4(b) we investigate the impact of the different clusters on Blockchain's literature. Cybersecurity, which has been one of the clusters that grew the most among all STEM clusters from the 1980s to the 2010s, has been stably the most influential cluster on Blockchain. The other top positions have instead changed from the first period considered, with Smart grids, which did not even have any influence on Blockchain at first, and Social network analysis becoming the most important clusters after Cybersecurity. Notice also how Robot swarms and Investments have experienced an increase in knowledge flow towards Blockchain, while the opposite has happened for Telecommunication, Optimization, Governance and Health.

Comparing the two plots in Fig. 4, we find examples of only unidirectional influences between Blockchain and the other clusters. The cluster of Social network analysis, third in position since 2015 to influence Blockchain, has not been influenced by it during the same period, which is also the case of Robot swarms and Governance. A similar situation is found for Wireless technologies, that has been strongly influenced by Blockchain over time, but only in recent years it has had a small impact on it.

Finally, we have conducted a similar analysis on the Governance cluster in Fig. S10 in the SM. In this case, we consider three different periods of times: 1950–1980, that is the early stage when it was the most important cluster overall; 1981-1990, when the amount of knowledge flow from Governance stopped to increase, still remaining among the top in terms of ranking; and 1991–2000, in which its role diminished and got surpassed by almost all other clusters by the end of the period. We do not find many noticeable differences between the first two periods. Most clusters have no significant knowledge flow from and to Governance, showing how *(de)centralization* developed independently in this cluster at first. Differently from Blockchain, the top clusters to have interactions with governance are Environment, Social network analysis and Investments. Wireless technologies, Blockchain and Robot swarms have also been influenced by Governance, but not vice-versa, apart from the sporadic case of Wireless technologies in the middle period. We can also see that the influence from Governance has increased over time on clusters like Blockchain, Optimization and Robot swarms, showing how the last years of the last century have been important milestones for the future of these clusters.

4 Discussion

In this paper we have developed and presented a framework that allowed us to quantitatively investigate how different topics have risen in the *(de)centralization* literature and have influenced it. By exploiting the S2AG corpus, we have shown that the literature on *(de)centralization* has exponentially increased in the past 70 years, with an author in 154 contributing to articles on the topic in 2021. We have observed